

Module 5

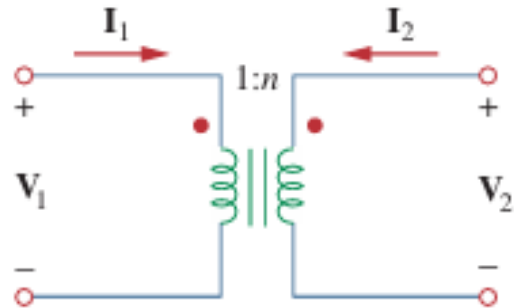
Two-Port Networks and Network Graphs

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Admittance parameters (y-parameters)

Impedance parameters may not exist for some two-port network. For example, consider the ideal transformer. The defining equations for this two-port network are:



$$V_1 = \frac{1}{n}V_2, \quad I_1 = -nI_2$$

So, there is a need for an alternative means of describing such a network. This need is met by the second set of parameters, which we obtain **by expressing the terminal currents in terms of the terminal voltages.**

In the y parameters, the terminal currents can be related to the terminal voltages as:

$$\begin{aligned} I_1 &= y_{11}V_1 + y_{12}V_2 \\ I_2 &= y_{21}V_1 + y_{22}V_2 \end{aligned}$$

where the y terms are called the admittance parameters, or simply y parameters, and have units of seimens.

or in matrix form as

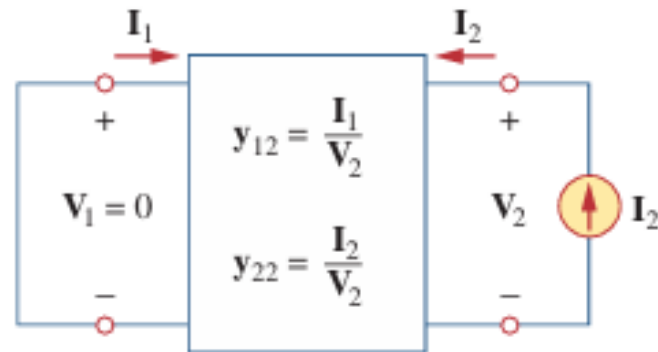
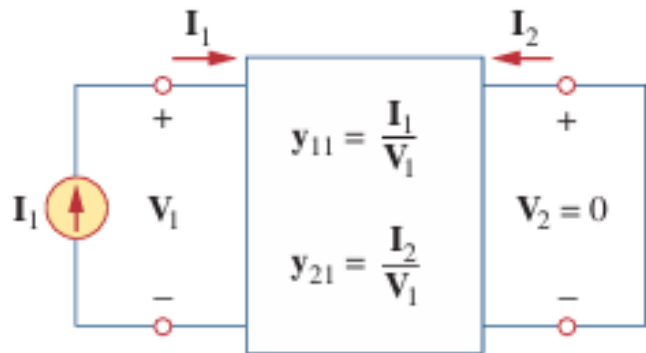
$$\begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = [y] \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

y- parameters

The values of the **y parameters** can be evaluated by setting (input port short-circuited) or (output port short-circuited). Thus,

$$\begin{aligned} y_{11} &= \left. \frac{I_1}{V_1} \right|_{V_2=0}, & y_{12} &= \left. \frac{I_1}{V_2} \right|_{V_1=0} \\ y_{21} &= \left. \frac{I_2}{V_1} \right|_{V_2=0}, & y_{22} &= \left. \frac{I_2}{V_2} \right|_{V_1=0} \end{aligned}$$

Since the y parameters are obtained by short-circuiting the input or out put port, they are also called the **short-circuit admittance parameters**.



y_{11} = Short-circuit input admittance

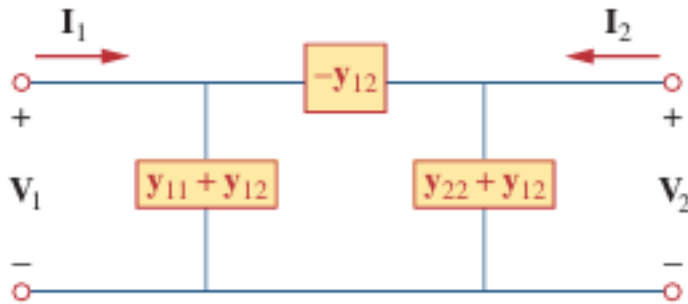
y_{12} = Short-circuit transfer admittance from port 2 to port 1

y_{21} = Short-circuit transfer admittance from port 1 to port 2

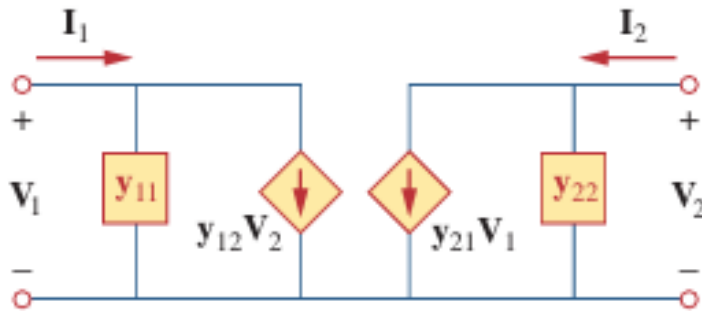
y_{22} = Short-circuit output admittance

y - parameters

For a two-port network that is **linear and has no dependent sources**, the transfer admittances are equal ($y_{12} = y_{21}$). This can be proved in the same way as for the z parameters. A reciprocal network can be modeled by the π -equivalent circuit.

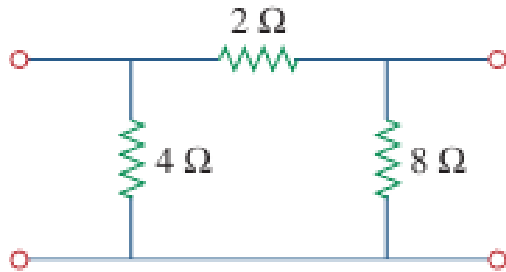


If the network is not reciprocal, a more general equivalent network is

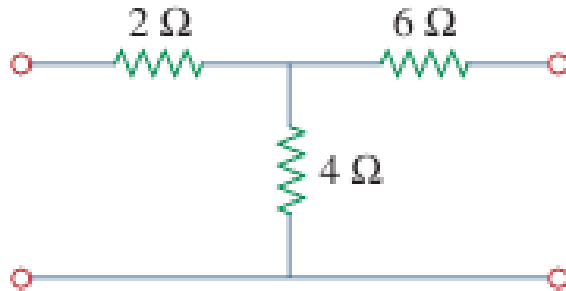


y - parameters

1. Obtain the y parameters for the network shown.

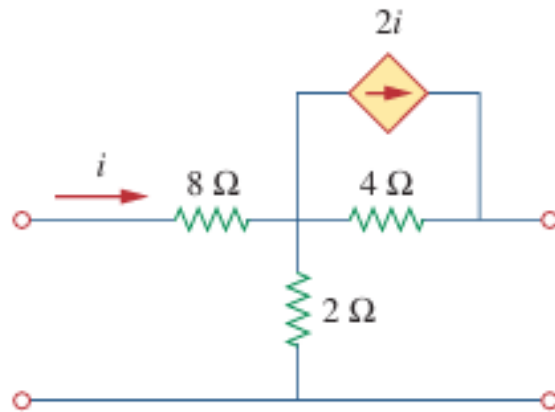


2 Obtain the y parameters for the T network shown.

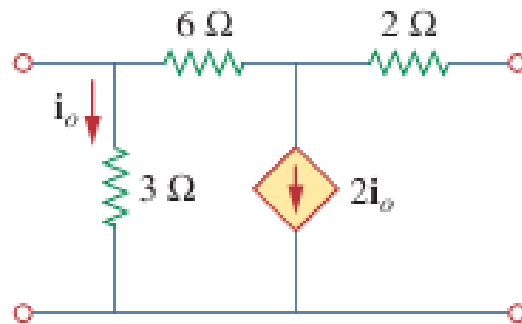


y - parameters

3. Determine the y parameters for the two-port shown.

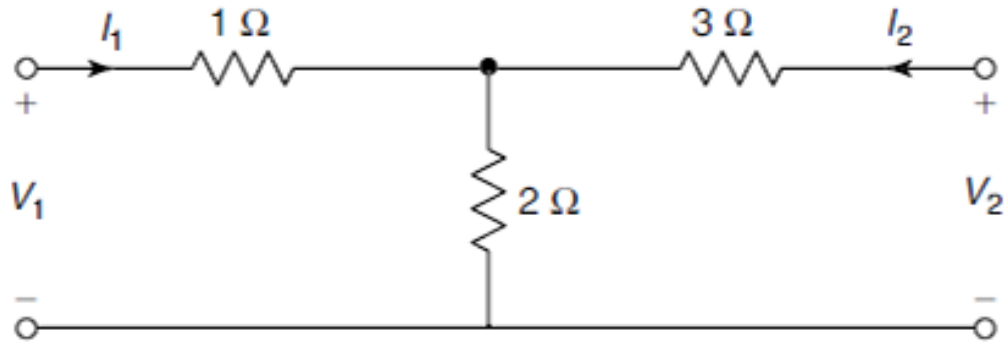


4. Obtain the y parameters for the circuit .

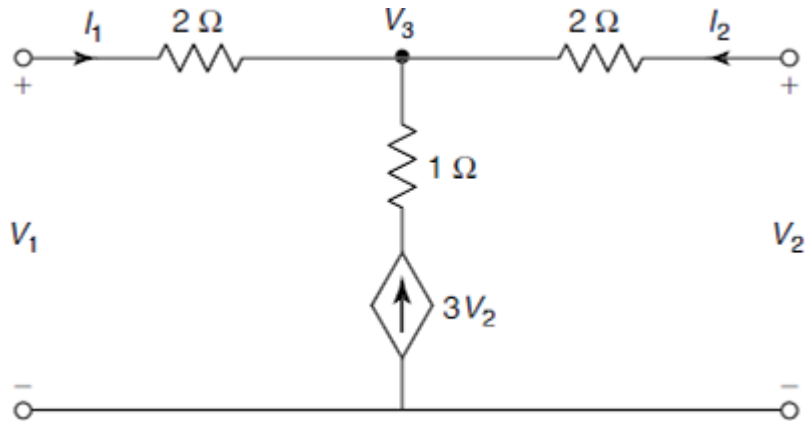


y - parameters

5. Find Y-parameters for the network shown.



6. Find y-parameters for the network shown.



Next class: Hybrid parameters